Example 4.28 If radii of two concentric circles are 4 cm and 5 cm then find the length of the chord of one circle which is a tangent to the other circle.

Solution $OA = 4$ cm, OB $= 5$ cm; also $OA \perp BC$.

$$OB^2 = OA^2 + AB^2$$

$$5^2 = 4^2 + AB^2 \text{ gives } AB^2 = 9$$

Therefore $AB = 3$ cm

$BC = 2$AB hence $BC = 2 \times 3 = 6$ cm

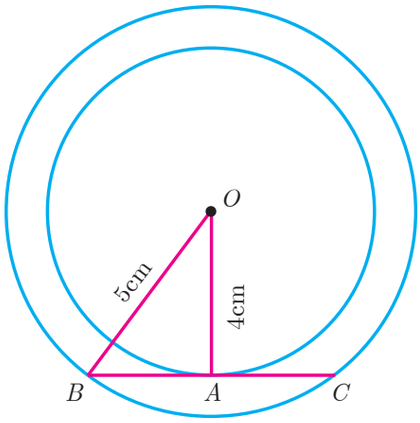


Fig. 4.66

4.5.1 Construction

Construction of tangents to a circle

Now let us discuss how to draw

(i) a tangent to a circle using its centre

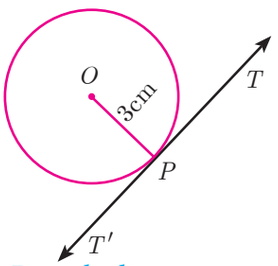
(ii) a tangent to a circle using alternate segment theorem

(iii) pair of tangents from an external point

Construction of a tangent to a circle (Using the centre)

Example 4.29 Draw a circle of radius 3 cm. Take a point P on this circle and draw a tangent at P.

Solution Given, radius $r = 3$ cm



Rough diagram

Construction

Step 1: Draw a circle with centre at O of radius 3 cm.

Step 2: Take a point P on the circle. Join OP.

Step 3: Draw perpendicular line to OP which passes through P.

Step 4: TT' is the required tangent.

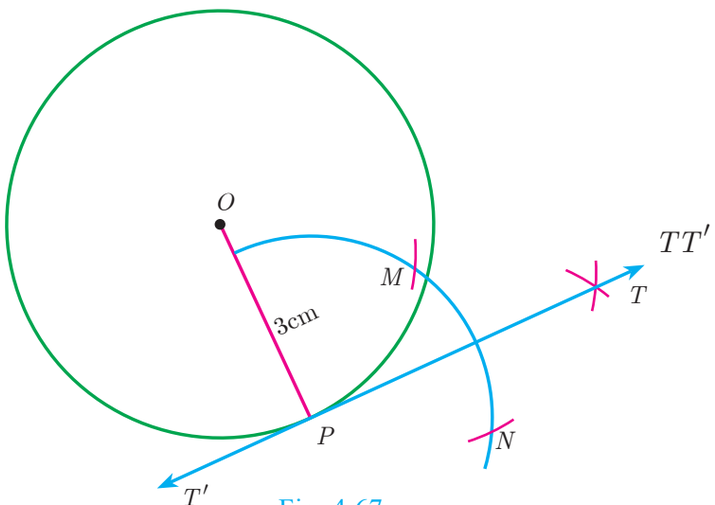


Fig. 4.67

Construct of a tangent to a circle (Using alternate segment theorem)

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